

6(a)	The magnetic flux density at a point in space is the magnetic <u>force per unit length</u> <u>per unit current</u> acting on a long straight conductor <u>carrying current and placed at right angle to the field</u> at that point. 1 marks for any 2 underlined phrases 2 marks for 3 underlined phrases	B2
(b)	Spacing between <u>circles</u> increases with distance from wire (at least three circles needed) Arrows showing direction of field is clockwise	B1 B1
(c)(i)	(each) wire lies/sits in the (magnetic) field generated/created by the other OR magnetic fields generated/created by the wires interact with each other. current (in one wire) is <u>perpendicular</u> (or not parallel) to (magnetic) field (due to other wire) so (magnetic) force acts (on wire)	B1 B1
(c)(ii)	Arrow drawn to the left of X, labelled F	B1
(c)(iii)	B-field on X due to Y = $\frac{\mu_0 I}{2\pi d} = \frac{4\pi \times 10^{-7} (3)}{2\pi (0.12)}$ = $5.0 \times 10^{-6} \text{ T}$ Force on X per unit length, $F = BIL$ $\frac{F}{L} = 5 \times 10^{-6} (2)$ = 10×10^{-6}	C1 A1
(c)(iv)	Magnetic field acting on Y by X is parallel to the wire Y. No magnetic force acts on Y.	A1

